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Deep recurrent Gaussian processes for outlier-robust system identification

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ABSTRACT

Gaussian Processes (GP) comprise a powerful kernel-based machine learning paradigm which has recently attracted the attention of the nonlinear system identification community, specially due to its inherent Bayesian-style treatment of the uncertainty. However, since standard GP models assume a Gaussian distribution for the observation noise, i.e., a Gaussian likelihood, the learning and predictive capabilities of such models can be severely degraded when outliers are present in the data. In this paper, motivated by our previous work on GP learning with data containing outliers and recent advances in hierarchical (deep GPs) and recurrent GP (RGP) approaches, we introduce an outlier-robust recurrent GP model, the RGP-*t*. Our approach explicitly models the observation layer, which includes a heavy-tailed Student-*t* likelihood, and allows for a hierarchy of multiple transition layers to learn the system dynamics directly from estimation data contaminated by outliers. In addition, we modify the original variational framework of standard RGP in order to perform inference with the new RGP-*t* model. The proposed approach is comprehensively evaluated using six artificial benchmarks, within several outlier contamination levels, and two datasets related to process industry systems (pH neutralization and heat exchanger), whose estimation data undergo large contamination rates. The simulation results obtained by the RGP-*t* model indicates an impressive resilience to outliers and a superior capability to learn nonlinear dynamics directly from highly outlier-contaminated data in comparison to existing GP models.

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1. Introduction

Nonlinear system identification, while of fundamental importance for the design and analysis of industrial processes, is a challenging task. Since the quality of the model is usually the bottleneck of the final problem solution [1], complications such as estimation data contaminated with outliers must be carefully considered [2]. Such scenario is very often encountered in everyday practice and is the focus of the present paper. For this purpose, we follow a Bayesian approach to system identification, where the model accounts for the uncertainty in the noisy data and in the learned dynamics [3]. More specifically, we apply Gaussian Processes (GP) models, a principled probabilistic approach to learning in kernel machines [4].

Although GPs have been used for predictions by the geostatistics community for many decades, where it is usually known as *kriging* [5,6], it was only some years later that works such as [7] and [8] indicated that GP models are capable to outperform conventional nonlinear regression frameworks, such as artificial neural networks (ANNs), support vector regression (SVR) and spline methods. GP models are nonparametric data-driven techniques, where instead of a rigid prespecified structure, the model allows for the data to "speak by itself". A typical GP model increases its complexity as more data becomes available and estimation data is used to both optimize the model and make predictions.

Another major distinctive feature of the GP framework over those powerful regression methods is that the outputs of a GP model are probabilistic distributions, i.e., instead of point estimates, each prediction is given by a fully defined distribution. As a consequence, besides computing mean values for predictions, one can easily compute as well prediction intervals associated to those

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values, using for instance the standard deviation of the predictive distribution.

Because of the aforementioned appealing features, application of GP models in engineering is gaining momentum. In the context of dynamical system modeling and identification, several authors have applied GP models to, for instance, local approaches for system identification [9,10], system identification with derivative observations [11], nonlinear autoregressive identification [12], learning nonstationary dynamics [13], multi-step ahead prediction with propagation of uncertainty [14,15], transfer function estimation [16] and state space modeling [17,18].

In standard applications of GP models a Gaussian observation noise is usually assumed, and hence Gaussian likelihoods arise naturally within this framework. While this is a reasonable assumption for many applications, it is not the case for modeling scenarios contaminated with non-Gaussian noise, such as impulsive noise.[1] In such scenarios the model's performance considerably deteriorates, compromising its out-of-sample prediction capability.

In the current paper, we pursue GP models that are able to directly learn dynamics from data containing outliers, as opposed to methodologies which remove them from the data before training, by promoting a *data cleaning* step [19]. Such an approach, surveyed in the context of temporal data in [20], can be useful when just a few outliers of high magnitude are present, but it becomes problematic when dealing with outliers [21] which are intermingled with the samples of the system under analysis, or when the outliers have their own "dynamics" apart from that of the system of interest. Huber [22], a pioneer in robust statistics, also argues against the data cleaning approach, emphasizing that dealing with outliers directly avoids erroneous removal of potentially valuable training samples.

Alternatively, robust GP-based modeling approaches for nonlinear regression usually apply heavy-tailed distributions for the model likelihood [23], such as Student-*t* [24–26], Laplace [27], and a mixture of Gaussians [28–30] in order to handle outliers as non-Gaussian noise. However, while the inference process (i.e., learning) in GP models with Gaussian likelihood is tractable,[2] non-Gaussian likelihood models often lead to non-tractable expressions, requiring either stochastic sampling techniques, such as Markov Chain Monte Carlo (MCMC) and Sequential Monte Carlo (SMC) [24,31–33], or deterministic approximation methods, such as variational Bayes (VB) [34–36] and expectation propagation (EP) [37,38].

Although sampling-based algorithms have been extensively used for nonlinear system identification in different scenarios [39–41], in this work we focus on variational approaches, widely used alternatives to approximate posterior probability densities with an optimization-based formulation which tends to be faster and scale up better to large datasets [42,43]. It is worth mentioning that variational methods have been largely used in several non-analytical GP variants such as sparse approximations [44], non-supervised learning [45], state-space models [17] and general hierarchical modeling [46].

Despite the increasing interest in GP-based modeling for outlier-robust nonlinear regression, developing such models specifically for robust dynamical system identification is a relatively new topic. In fact, we were able to find very few contributions other than our own previous work [47,48]. In [32], the authors propose a robust GP model for estimating impulse responses of linear system when the measurements are corrupted by outliers, without addressing nonlinear systems. More recently, in [49] the authors introduce an Expectation-Maximization (EM) algorithm to tackle robust regression tasks, reporting an experiment in system identification, which was evaluated in one-step-ahead prediction scenarios only.

In [47] two robust GP models were applied for dynamical system identification using NARX (nonlinear autoregressive with exogenous inputs) models: a GP model with Student-*t* likelihood and variational inference (GP-tVB) and a GP model with Laplace likelihood and EP inference (GP-LEP). Both models outperformed the standard GP-NARX with Gaussian likelihood, especially GP-tVB, but they were still very sensitive to outliers in some of the studied evaluation scenarios.

A novel robust latent autoregressive model, named GP-RLARX, was proposed by us in [48]. This model incorporated the Student-*t* likelihood and introduced a latent dynamical structure to account for the uncertainty in the regressors, avoiding the explicit feedback of the noisy outputs in the model dynamics. The GP-RLARX followed a variational framework for inference and performed well in several artificial benchmarks with different levels of outlier contamination.

Despite its superior performance in the presence of outliers when compared to GP-tVB, the GP-RLARX model adopts a very simple observation model: the output is equal to the most recent latent variable plus noise. This forces the latent space to be closely related to the output and constrains the latent variables of the model. Furthermore, it relies on a single transition layer. As argued in [50], a hierarchical (or *deep*) structure can be helpful for recurrent modeling. Besides, such multilayer structure has been successfully exploited more recently with the rise of deep GP models [51,46].

The aforementioned restrictions could hinder the GP-RLARX model's capability to learn more complex dynamics from outlier-corrupted data. Bearing this in mind, in the current paper we extend GP-RLARX by including an additional GP to model the observation (or emission) layer, in order to separate the transition and emission nonlinear functions. To further increase the representational capability of the model, we allow the inclusion of more than one transition layer, referred to as *hidden layers*. This approach builds upon our work on Recurrent Gaussian Processes (RGP) model presented in [52], being equivalent to provide it with a Student-*t* likelihood. However, such extension requires a modification in the Recurrent Variational Bayes (REVARB) framework described in [52], since it originally considered a Gaussian likelihood.

In summary, the proposed extension aims to bring together in a synergistic way the best properties of the two previous approaches in order to obtain eventually a more reliable solution for robust system identification: (*i*) *resilience to non-Gaussian noise* provided by the GP-RLARX model due to the Student-*t* likelihood, and (*ii*) the enhanced *representational capability* provided by the hierarchical RGP structure. We name henceforth this new model as the RGP-*t* model and the corresponding modified variational framework as REVARB-*t*, whose details will be presented later on this paper.

We evaluate the proposed RGP-*t* model in system identification tasks with outliers using for that purpose several artificial benchmarks and datasets related to industrial systems. The impressive results obtained by the RGP-*t* model indicate that its powerful multilayer structure is able to learn the system dynamics from data contaminated with non-Gaussian noise and to perform simulation on unseen data better than previous robust GP models. Furthermore, we also illustrate how it can be used not only to deal with outliers (by not being influenced by them), but also to successfully detect them in the estimation data.

The remainder of the paper is organized as follows. Section 2 summarizes the standard GP modeling framework, highlighting how it can be directly applied to NARX models. In Section 3 we get to the proposed RGP-*t* model from the original GP-RLARX and RGP formulations. In Section 4 we detail REVARB-*t*, the modified variational framework that enables robust inference with RGP-*t*. In Section 5 we report and analyze the results obtained with RGP-*t*

[1] Commonly treated as a type of outlier by standard Gaussian models.

[2] A mathematical expression is considered tractable if its solution is analytical and, hence, do not require the use of approximate methods.

on robust system identification tasks. We conclude the paper in Section 6.

2. The GP modeling framework

Following a probabilistic modeling approach, the GP framework defines a prior distribution over functions. By imagining functions as infinitely long vectors, all points generated by some unknown function $f(\cdot)$ are jointly modeled as a single GP, an infinite object. Fortunately, because of the marginalization property of Gaussians,[3] we can analyze such object by working with a finite multivariate Gaussian distribution over the vector $\boldsymbol{f} = f(\boldsymbol{X}) \in \mathbb{R}^N$, given the inputs $\boldsymbol{x}_i \in \mathbb{R}^D, \forall i \in \{1, \ldots, N\}$, stacked in the matrix $\boldsymbol{X} \in \mathbb{R}^{N \times D}$, such as [4]:

$$p(\boldsymbol{f}|\boldsymbol{X}) = \mathcal{N}(\boldsymbol{f}|\boldsymbol{0}, \boldsymbol{K}_f). \tag{1}$$

The prior distribution in Eq. (1) is fully characterized by its zero mean vector $\boldsymbol{0} \in \mathbb{R}^N$ and covariance matrix $\boldsymbol{K}_f \in \mathbb{R}^{N \times N}$, whose elements are calculated by $[\boldsymbol{K}_f]_{ij} = k(\boldsymbol{x}_i, \boldsymbol{x}_j), \forall i, j \in \{1, \ldots, N\}$, where $[\cdot]_{ij}$ indicates the element in the ith row and jth column of a matrix. The so-called covariance (or *kernel*) function $k(\cdot, \cdot)$ is restricted so that it generates a positive semidefinite covariance matrix $\boldsymbol{K}_f$, which in GP models is also called *kernel matrix*. The exponentiated quadratic kernel is a common choice and will be applied throughout this work:

$$k(\boldsymbol{x}_i, \boldsymbol{x}_j) = \sigma_f^2 \exp\left[-\frac{1}{2}\sum_{d=1}^{D} w_d^2 (x_{id} - x_{jd})^2\right], \quad \forall i, j \in \{1, \ldots, N\}, \tag{2}$$

where x_{id} and x_{jd} are respectively the dth components of the inputs $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$. The kernel hyperparameters can be gathered into the vector $\boldsymbol{\theta} = [\sigma_f^2, w_1^2, \ldots, w_D^2]^\top$ and are responsible for characterizing the covariance of the model [4].

The true values of $\boldsymbol{f}$ are usually not available and the regression must be performed instead with the observations $y_i = f(\boldsymbol{x}_i) + \epsilon_i^{(y)}$, contaminated with some noise $\epsilon_i^{(y)}$. If we consider the observation noise to be independent and Gaussian with zero mean and variance σ_y^2, we obtain the likelihood $p(\boldsymbol{y}|\boldsymbol{f}) = \mathcal{N}(\boldsymbol{y}|\boldsymbol{f}, \sigma_y^2\boldsymbol{I})$, where $\boldsymbol{y} \in \mathbb{R}^N$ is the vector of noisy observations and $\boldsymbol{I} \in \mathbb{R}^{N \times N}$ is the identity matrix. The noise variance σ_y^2 is usually treated as a hyperparameter and included in the hyperparameters vector $\boldsymbol{\theta}$.

In such noisy regression scenario, the marginal likelihood $p(\boldsymbol{y}|\boldsymbol{X})$ of the observed data is obtained after marginalizing the variables $\boldsymbol{f}$, which is analytical, since both the likelihood $p(\boldsymbol{y}|\boldsymbol{f})$ and the prior $p(\boldsymbol{f}|\boldsymbol{X})$ are Gaussian [4,43]:

$$\begin{aligned} p(\boldsymbol{y}|\boldsymbol{X}) &= \int_{\boldsymbol{f}} p(\boldsymbol{y}|\boldsymbol{f}) p(\boldsymbol{f}|\boldsymbol{X}) = \int_{\boldsymbol{f}} \mathcal{N}(\boldsymbol{y}|\boldsymbol{f}, \sigma_y^2\boldsymbol{I}) \mathcal{N}(\boldsymbol{f}|\boldsymbol{0}, \boldsymbol{K}_f) \\ &= \mathcal{N}(\boldsymbol{y}|\boldsymbol{0}, \boldsymbol{K}_f + \sigma_y^2\boldsymbol{I}), \end{aligned} \tag{3}$$

where the subindex in the integral symbol indicates the integrated variables,[4] i.e., $\int_{\boldsymbol{f}}$ is equivalent to $\int_{-\infty}^{\infty} \mathrm{d}\boldsymbol{f}$. It should be noted that in order to avoid notation clutter we have omitted the conditioning on the hyperparameters vector $\boldsymbol{\theta}$.

[3] Given a collection of jointly Gaussian variables, the joint marginal distribution of any subset of such variables is also Gaussian.

[4] Such compact notation for the marginalization of a random variable is used, for instance, in Barber's book [42]. Note that the integral covers all possible values taken by the integrated variables.

Given a new input $\boldsymbol{x}_*$, the posterior predictive distribution of the related output f_* is calculated analytically using standard Gaussian distribution conditioning properties:

$$p(f_*|\boldsymbol{y}, \boldsymbol{X}, \boldsymbol{x}_*) = \mathcal{N}(f_*|\mu_*, \sigma_*^2),$$

which holds given the zero-mean joint Gaussian distribution below:

$$p(\boldsymbol{y}, f_*) = \mathcal{N}\left(\begin{bmatrix}\boldsymbol{y} \\ f_*\end{bmatrix} | \boldsymbol{0}, \begin{bmatrix}\boldsymbol{K}_f + \sigma_y^2\boldsymbol{I} & \boldsymbol{k}_{f*} \\ \boldsymbol{k}_{*f} & k_{**}\end{bmatrix}\right), \tag{5}$$

where $\boldsymbol{k}_{*f} = [k(\boldsymbol{x}_*, \boldsymbol{x}_1), \ldots, k(\boldsymbol{x}_*, \boldsymbol{x}_N)]$, $\boldsymbol{k}_{f*} = \boldsymbol{k}_{*f}^\top$ and $k_{**} = k(\boldsymbol{x}_*, \boldsymbol{x}_*)$. The predictive distribution of y_* is similar to the one in Eq. (4), but the variance is added by σ_y^2.

Model selection consists in finding the *maximum likelihood*[5] (ML) solution $\boldsymbol{\theta}_{\mathrm{ML}}$ for the hyperparameters vector, which is performed by maximizing the marginal log-likelihood $\log p(\boldsymbol{y}|\boldsymbol{X})$ of the observed data, the so-called *evidence* of the model:

$$\begin{aligned} &\boldsymbol{\theta}_{\mathrm{ML}} = arg\max_{\boldsymbol{\theta}} \log p(\boldsymbol{y}|\boldsymbol{X}), \\ &\log p(\boldsymbol{y}|\boldsymbol{X}) = \log \mathcal{N}(\boldsymbol{y}|\boldsymbol{0}, \boldsymbol{K}_f + \sigma_y^2\boldsymbol{I}), \\ &= -\frac{N}{2}\log 2\pi - \frac{1}{2}\log|\boldsymbol{K}_f + \sigma_y^2\boldsymbol{I}| - \frac{1}{2}\boldsymbol{y}^\top\left(\boldsymbol{K}_f + \sigma_y^2\boldsymbol{I}\right)^{-1}\boldsymbol{y}, \end{aligned} \tag{6}$$

where $|\cdot|$ denotes the determinant of a matrix. Optimization is guided by the analytical gradients of the evidence with respect to the hyperparameters vector $\boldsymbol{\theta}$. Such model selection step aims to find an appropriate GP model that explains the data. After the optimization, the hyperparameters $w_1^2, \ldots, w_D^2$, taking values close to zero implement *automatic relevance determination* (ARD), by turning off the unrelevant components of the input vector. Algorithm 1 summarizes the use of the standard GP modeling framework for regression.

Remark 1. The matrix $\left(\boldsymbol{K}_f + \sigma_y^2\boldsymbol{I}\right)$ that appears in the GP expressions is similar to regularized kernel methods. However, the hyperparameter related to the regularization of the matrix $\boldsymbol{K}_f$, the noise variance σ_y^2, is estimated directly from the data by optimization of the model evidence in Eq. (6). No random or grid search procedures, common for instance in the SVR framework, are required.

Algorithm 1. Standard GP modeling for regression.

- Estimation step
Require: $\boldsymbol{X} \in \mathbb{R}^{N \times D}$ (inputs), $\boldsymbol{y} \in \mathbb{R}^N$ (outputs)
Initialize kernel hyperparameters $\boldsymbol{\theta} = \left[\sigma_f^2, w_1^2, \ldots, w_D^2, \sigma_n^2\right]^\top$;
repeat
Compute the model evidence $\log p(\boldsymbol{y}|\boldsymbol{X})$ via Eq. (6).
Compute the analytical gradients $\frac{\partial \log p(\boldsymbol{y}|\boldsymbol{X})}{\partial \boldsymbol{\theta}}$ of the Eq. (6) with respect to the kernel hyperparameters;
Update the hyperparameters $\boldsymbol{\theta}$ with a gradient-based method (e.g. BFGS);
until convergence or maximum number or iterations
Output the optimized hyperparameters $\boldsymbol{\theta}_{ML}$;
- Test step
Require: $\boldsymbol{x}_* \in \mathbb{R}^D$ (inputs), $\boldsymbol{\theta}_{ML}$
Compute the predictive mean μ_* and variance σ_*^2 (Eq. (4));
Output $f_* \sim \mathcal{N}(\mu_*, \sigma_*^2)$;

2.1. GP-NARX for dynamical system identification

In [12] the authors discuss how the GP formalism can be directly applied to NARX models with Gaussian observation noise. In the time step i, the input becomes the regressor vector $\boldsymbol{x}_i = [y_{i-1}, \ldots, y_{i-L}, u_{i-1}, \ldots, u_{i-L_u}]^\top$ comprised of L past observed outputs $y_i \in \mathbb{R}$ and L_u past exogenous inputs $u_i \in \mathbb{R}$. Model selection,

[5] In this context also called *type II maximum likelihood* [4].

i.e., hyperparameters optimization, and inference then follows the standard GP regression model case.

In what concerns the validation of the identified model, the following remark is timely.

Remark 2. In literature, many authors still validate the identified models using one-step-ahead predictions, where previous inputs and observed outputs until the current time step are used as regressors to predict the output in the next time step. However, this validation methodology can be misleading, because even "poor models can look good" [53]. Thus, we follow the more adequate procedure where only given inputs and past estimated outputs are used for prediction, which is called *infinite step ahead prediction*, *model predicted output*, *free simulation* or simply *simulation*. Such validation procedure more rigorously verify if the system's dynamics were properly identified.

3. From GP-RLARX to RGP-*t*: a robust recurrent GP model

As mentioned in the previous section, the GP modeling equations are tractable only when a Gaussian observation noise is considered. It is widely known that the light tails of the Gaussian likelihood do not model appropriately impulsive noise or other non-Gaussian observations. Hence, when dealing with outliers, we shall consider heavy tailed distributions, which we hypothesize that can suitably cope with non-Gaussian noise in the data. Additionally, if autoregressive models are chosen with the aim of capturing the system dynamics, the presence of outliers becomes even more problematic, since the predicted noisy outputs are inserted back into the input regressors, greatly interfering with the model capability to learn the underlying dynamics of the system.

The GP-RLARX model that we proposed in [48] as a robust alternative to the GP-NARX approach, besides incorporating a Student-*t* likelihood to account for outliers in the estimation data, also introduces a latent autoregressive structure to avoid outliers being directly used as input regressors.

The observation model of GP-RLARX follows the Student-*t* treatment detailed by Kuss in [27] in his thesis (and previously used in [54]), where a GP model for robust regression is described. We name that model, which does not incorporate dynamical latent variables, the GP-tVB model.

Instead of directly applying the Student-*t* probability density function in the likelihood, both GP-tVB and GP-RLARX models exploit the fact that it can be defined as a mixture of infinitely many Gaussian distributions with gamma distributed precisions (inverse variances)[6] [27]:

$$\mathcal{T}(\epsilon|\nu,\mu=0,\sigma)=\frac{\Gamma((\nu+1)/2)}{\Gamma(\nu/2)\sqrt{\pi\nu\sigma^2}}\left(1+\frac{\epsilon^2}{\nu\sigma^2}\right)^{-(\nu+1)/2}, \tag{7}$$

$$\mathcal{T}(\epsilon|\nu=2\alpha,\mu=0,\sigma=\sqrt{\beta/\alpha})=\int_0^\infty \mathcal{N}(\epsilon|0,\tau^{-1})\Gamma(\tau|\alpha,\beta)\,\mathrm{d}\tau, \tag{8}$$

where zero mean Gaussians with τ precision were considered and α and β are respectively the shape and inverse scale hyperparameters of the gamma distribution. The resulting Student-*t* distribution for the variable ϵ, which also has zero mean, is characterized by the degrees of freedom ν and the scale hyperparameter σ. Lower values of ν result in heavier tails, while for $\nu\to\infty$ the tails become lighter and the distribution converges to a Gaussian.

[6] In [27] an inverse gamma prior was chosen for the variance of the Gaussian distribution, which is equivalent to the presented analysis.

Thus, the GP-tVB model can be described by the NARX formulation below [47]:

$$y_i=f(y_{i-1},\ldots,y_{i-L},u_{i-1},\ldots,u_{i-L_u})+\epsilon_i^{(y)}, \tag{9}$$

$$\boldsymbol{f}\sim\mathcal{N}\left(\boldsymbol{0},\boldsymbol{K}_f\right),\ \epsilon_i^{(y)}\sim\mathcal{N}\left(0,\tau_i^{-1}\right),\ \tau_i\sim\Gamma(\alpha,\beta), \tag{10}$$

where $f(\cdot)$ has a GP prior, L is the number of considered past outputs and the observation noise $\epsilon_i^{(y)}$ follows a Student-*t* distribution written using the aforementioned mixture approach, i.e., it is considered to be sampled from a Gaussian with gamma distributed precision. It is worth mentioning that, although the precisions τ_i follow the same gamma prior, each noise sample $\epsilon_i^{(y)}$ has its own precision value.

Following the same strategy for the likelihood and including a latent autoregressive structure, the GP-RLARX model defined in [48] is given by

$$x_i=f(x_{i-1},\ldots,x_{i-L},u_{i-1},\ldots,u_{i-L_u})+\epsilon_i^{(x)},$$

where L now is the number of considered past latent variables, $\epsilon_i^{(x)}\sim\mathcal{N}(0,\sigma_x^2)$ is a zero mean Gaussian transition noise with variance σ_x^2 and the observation noise $\epsilon_i^{(y)}$ is again Student-*t* distributed. It should be noted the absence of past observations in the regressors presented in Eq. (11).

We can see in Eq.(12) that the observation function is not separately modeled by the GP-RLARX, since y_i is simply given by the latent variable x_i plus the noise $\epsilon_i^{(y)}$. Thus, the variable x_i is driven close to y_i during estimation, which constrains its value and limits the representational power of the latent space. Besides, the function $f(\cdot)$ in Eq.(11) is the only modeled function, which does not support any kind of hierarchy.

By its turn, the deep and recurrent structure of the RGP model proposed by us in [52] generalizes the latent autoregressive structure from GP-RLARX by adding a GP prior to the observation function and allowing any number of hidden transition layers, further increasing the model's representational power. However, the RGP model originally uses a standard Gaussian observation likelihood, which makes it not suitable to scenarios where outliers are expected to occur.

From the exposed so far, in the present work we aim at combining the powerful recurrent and multilayer structure of RGPs with the robustness to outliers of the GP-RLARX to propose the RGP-*t* model. Considering H hidden transition layers and one observation (or emission) layer, the model is characterized by the following equations:

$$x_i^{(h)}=f^{(h)}\left(\hat{\boldsymbol{x}}_i^{(h)}\right)+\epsilon_i^{(h)},\ 1\le h\le H,$$

Each layer h is related to an unknown nonlinear function $f^{(h)}(\cdot)$ modeled by a distinct GP with covariance matrix $\boldsymbol{K}_f^{(h)}$. We emphasize that, while the transition noises $\epsilon_i^{(h)}$ are Gaussian, the observation noise $\epsilon_i^{(H+1)}$ is Student-*t* and capable of handling outliers. Differently from the GP-RLARX model, the RGP-*t* models at least two nonlinear functions, when $H=1$, which corresponds to a hierarchy of at least two separate GP priors.

In the previous equations, each layer's input $\hat{\boldsymbol{x}}_i^{(h)}$ is a regressor vector given by [52]:

$$\hat{\boldsymbol{x}}_i^{(h)}=\begin{cases}\left[\bar{\boldsymbol{x}}_{i-1}^{(1)},\bar{\boldsymbol{u}}_{i-1}\right]^\top, & \text{if } h=1,\\ \left[\bar{\boldsymbol{x}}_{i-1}^{(h)},\bar{\boldsymbol{x}}_i^{(h-1)}\right]^\top, & \text{if } 1<h\le H,\\ \bar{\boldsymbol{x}}_i^{(H)}, & \text{if } h=H+1,\end{cases} \tag{19}$$

where $\bar{\boldsymbol{x}}_i^{(h)}=\left[x_{i-1}^{(h)},\ldots,x_{i-L}^{(h)}\right]$, and $\bar{\boldsymbol{u}}_i^{(h)}=\left[u_{i-1},\ldots,u_{i-L_u}\right]$.

We note that the vector $\bar{\boldsymbol{x}}_i^{(h)}$ represents the autoregressive state associated with the layer h in the instant i, from which the dynamics are learned. Importantly, none of the layers' inputs defined in Eq. (19) are possibly formed by outlier-corrupted observations $\boldsymbol{y}$. Since the vectors $\bar{\boldsymbol{x}}_i^{(h)}$ are not directly dependent on past observations, they are related to general latent states of regular state space models. Thus, in the nonlinear system identification terminology introduced by Nelles in [1], the RGP-t is considered a model with *internal* dynamics, as opposed, for instance, to NARX models, which present *external* dynamics.

In Fig. 1 we illustrate the structures of the outlier-robust dynamical GP-based models used in this work: the GP-tVB (adapted from [27]), the GP-RLARX [48] and the RGP-t (this work). In the graphical models, the shaded nodes are random observed entities (outputs), the white nodes represent the latent (unobserved) variables and the past exogenous inputs $\bar{\boldsymbol{u}}_{i-1}$ are treated as deterministic elements. The arrows indicate direct dependencies, e.g., the observation y_i depends on the noise $\epsilon_i^{(y)}$. We explicit the noise models and the latent f_i variables, even though the latter are always integrated out (i.e., marginalized). The models also include the hyperparameters α and β related to the gamma distributed precision of the observation noise. The variance of the Gaussian transition noises are not shown, but are implicit. Table 1 summarizes the features presented by the aforementioned approaches, comparing with the non-robust GP-NARX and RGP models. Note that the robust model used in [49] follows the same NARX structure of the GP-tVB, although with a different inference method (an EM algorithm).

Remark 3. The model formulation evolution from GP-tVB to GP-RLARX, and then to RGP-t, represents a pursuit towards more expressive models that are, at the same time, robust to outliers and tailored to nonlinear system identification. As can be inferred from Fig. 1, the GP-tVB is simply an adaptation of a robust regression model to the NARX structure. The GP-RLARX then introduces latent dynamical variables, which, although closely related to the observations, are properly handled as random variables. Finally, the proposed RGP-t enhances the previous approaches by incorporating a fully hierarchical structure, increasing its representational power while preserving the focus on simulation with uncertainty propagation along all the dynamical layers and a robust nonlinear observation layer.

The non-robust RGP model is trained with the Recurrent Variational Bayes (REVARB) framework, proposed in [52]. However, the standard REVARB considers only a Gaussian likelihood. In order to train the newly proposed RGP-t model, we also introduce in this paper a robust modification to REVARB, named REVARB-t and explained in the next section.

4. REVARB-t: a robust recurrent variational framework

The variational framework presented in [52] for the non-robust RGP model tackles the intractabilities related to the probabilistic treatment of the latent dynamical variables. However, the proposed RGP-t introduces the need to marginalize the variables $\boldsymbol{\tau}$ related to the noise precision in the observation layer in order to obtain the model marginal likelihood, as we have done in Section 2 for the standard GP model.

Thus, we have two goals in this section: (*i*) approximately marginalize all the latent variables of the RGP-t model, including the observation noise precisions $\boldsymbol{\tau}$, and (*ii*) find an expression that approximates the RGP-t model evidence, i.e., the marginal log-likelihood, which can be used to perform model selection. The extended REVARB-t framework, to be described henceforward, fulfills both goals and considers all specificities necessary to perform inference with the robust RGP-t model.

For the sake of clarity, we follow the steps of the original REVARB method described in [52], emphasizing the modifications that led to the proposal of REVARB-t. We first rewrite Eqs. (15) and (16) and explicit the involved distributions:

$$\begin{aligned}
p\left(\boldsymbol{f}^{(h)}|\hat{\boldsymbol{X}}^{(h)}\right) &= \mathcal{N}\left(\boldsymbol{f}^{(h)}|\boldsymbol{0}, \boldsymbol{K}_f^{(h)}\right), && \forall h \in \{1,\ldots,H+1\},\\
p\left(x_i^{(h)}\right) &= \mathcal{N}\left(x_i^{(h)}|\mu_{0i}^{(h)}, \lambda_{0i}^{(h)}\right), && \forall i \in \{1,\ldots,L\},\\
p\left(x_i^{(h)}|f_i^{(h)}\right) &= \mathcal{N}\left(x_i^{(h)}|f_i^{(h)}, \sigma_h^2\right), && \forall i \in \{L+1,\ldots,N\},\\
p\left(y_i|f_i^{(H+1)}, \tau_i\right) &= \mathcal{N}\left(y_i|f_i^{(H+1)}, \tau_i^{-1}\right), && \forall i \in \{L+1,\ldots,N\},\\
p(\tau_i) &= \Gamma(\tau_i|\alpha,\beta), && \forall i \in \{L+1,\ldots,N\},
\end{aligned}$$

where $\hat{\boldsymbol{X}}^{(h)}$ is simply obtained by stacking the vectors $\hat{\boldsymbol{x}}_i^{(h)}, \forall i \in \{L+1,\ldots,N\}$ and we have also made explicit the means $\mu_{0i}^{(h)}$ and variances $\lambda_{0i}^{(h)}$ of the initial L latent variables of each layer. Such set of distributions result in a prior generative model, since we could sequentially take samples from it even before observing any data. It is relevant to note that the presence of the gamma distribution, introduced by the new RGP-t formulation, would allow the generation of some high variance (small precision) noise samples, which account for the outliers in the data.

The joint distribution of all the variables present in the RGP-t model is given by

$$\begin{aligned}
&p\left(\boldsymbol{y}, \boldsymbol{\tau}, \left\{\boldsymbol{x}^{(h)}\right\}|_{h=1}^{H}, \left\{\boldsymbol{f}^{(h)}\right\}|_{h=1}^{H+1}\right) =\\
&\prod_{i=L+1}^{N} p\left(y_i|f_i^{(H+1)}, \tau_i\right) p(\tau_i) p\left(f_i^{(H+1)}|\hat{\boldsymbol{x}}_i^{(H)}\right)\\
&\left(\prod_{h=1}^{H} p\left(x_i^{(h)}|f_i^{(h)}\right) p\left(f_i^{(h)}|\hat{\boldsymbol{x}}_i^{(h)}\right)\right)\left(\prod_{i=1}^{L}\prod_{h=1}^{H} p\left(x_i^{(h)}\right)\right).
\end{aligned} \tag{20}$$

In order to obtain the marginal likelihood of the model, we would need to marginalize all the latent variables in the joint distribution in Eq. (20). However, such integration is intractable (non-analytical), due to the presence of the latent variables $\boldsymbol{x}_i^{(h)}$, which appear in the terms $\left|\boldsymbol{K}_f^{(h)}\right|$ and $\left(\boldsymbol{K}_f^{(h)}\right)^{-1}$ inside $p\left(\boldsymbol{f}^{(h)}|\boldsymbol{X}^{(h)}\right)$.

Similar intractabilities were tackled by Titsias' sparse variational framework [44]. Following this approach, we augment each layer h by including M inducing points $\boldsymbol{z}^{(h)} \in \mathbb{R}^M$ evaluated in M pseudo-inputs $\boldsymbol{\zeta}_j^{(h)} \in \mathbb{R}^{D_h}, \forall j \in \{1,\ldots,M\}$, where D_h is the dimension of the hth layer input $\hat{\boldsymbol{x}}_i^{(h)} \in \mathbb{R}^{D_h}$. The inducing points $\boldsymbol{z}^{(h)}$ are considered to be extra samples from the same GP that models $f^{(h)}(\cdot)$, i.e., $p(\boldsymbol{z}^{(h)}) = \mathcal{N}\left(\boldsymbol{z}^{(h)}|\boldsymbol{0}, \boldsymbol{K}_z^{(h)}\right)$, where $\boldsymbol{K}_z^{(h)} \in \mathbb{R}^{M\times M}$ is the covariance matrix obtained from the pseudo-inputs $\boldsymbol{\zeta}_j^{(h)}|_{j=1}^{M}$. Then, we replace in Eq. (20) the terms $p\left(f_i^{(h)}|\hat{\boldsymbol{x}}_i^{(h)}\right)$ with $p\left(f_i^{(h)}|\boldsymbol{z}^{(h)}, \hat{\boldsymbol{x}}_i^{(h)}\right) p(\boldsymbol{z}^{(h)})$. Such augmentation does not fundamentally change the original model, since if we integrate out $\boldsymbol{z}^{(h)}$ we recover the same original expression.

The exact marginal likelihood $p(\boldsymbol{y})$ still cannot be computed analytically, but we are now able to lower bound it by applying Jensen's inequality,[7] following standard variational methods [55]:

$$\log p(\boldsymbol{y}) \geq \int_{\boldsymbol{\tau},\boldsymbol{f},\boldsymbol{x},\boldsymbol{z}} Q$$

[7] In its probabilistic form, Jensen's inequality states that $\phi(\mathbb{E}\{x\}) \leq \mathbb{E}\{\phi(x)\}$, where $\phi(\cdot)$ is a convex function and $\mathbb{E}\{\cdot\}$ is the expectation operator. Since the function $\log(\cdot)$ is concave, the side of the inequality is changed in Eq. (21).

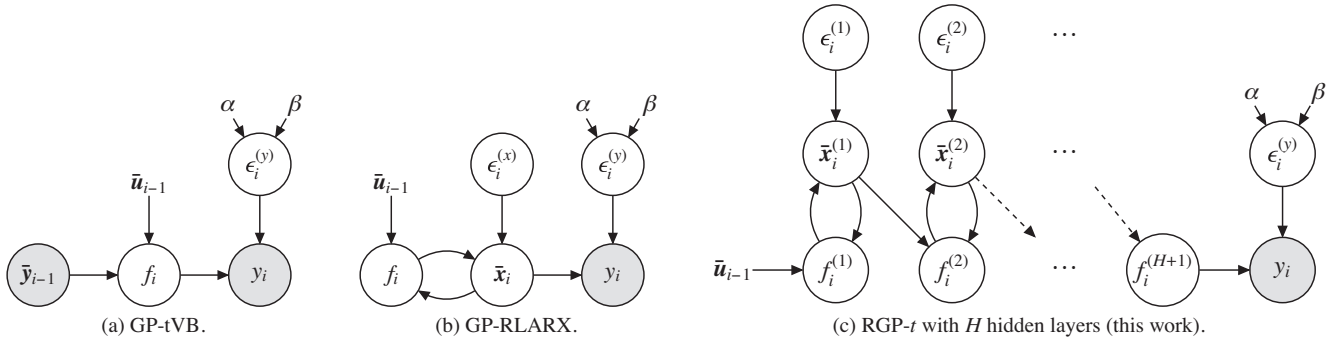


Fig. 1. Graphical models for the robust GP-based approaches used in this paper. The shaded nodes are random observed entities (outputs), while the white nodes represent the latent (unobserved) variables. The arrows indicate direct dependencies. It should be noted that past exogenous inputs $\bar{\boldsymbol{u}}_{i-1}$ are treated as deterministic elements.

Table 1
Summary of features presented by the different GP-based models used in this work for nonlinear system identification.

	Latent states	Hierarchical	Propagates uncertainty	Outlier-robust	Inference
GP-NARX					Analytical
GP-tVB				√	Variational
GP-RLARX	√		√	√	Variational
RGP	√	√	√		REVARB
RGP-t (this work)	√	√	√	√	REVARB-t

$$\log\left[\frac{p\left(\boldsymbol{y},\boldsymbol{\tau},\left\{\boldsymbol{x}^{(h)}\right\}|_{h=1}^{H},\left\{\boldsymbol{f}^{(h)},\boldsymbol{z}^{(h)}\right\}|_{h=1}^{H+1}\right)}{Q}\right], \tag{21}$$

where the indexes in the integral symbol indicate the integrated variables in all layers[8] and Q is the so-called variational distribution, which can be freely chosen and will be used to approximate the posterior of the unknown variables. The bound in Eq. (21) acts as a proxy for the log-marginal likelihood and by making it tighter, i.e., maximizing it, the approximation gets closer to the true expression for $\log p(\boldsymbol{y})$ [43,42]. Furthermore, it can be proven that such maximization is equivalent to minimizing the Kullback–Leibler divergence between the approximate variational posterior Q and the true posterior [35,36].

We conveniently choose the factorized form below for Q:

$$Q = q(\boldsymbol{\tau})\left(\prod_{h=1}^{H} q\left(\boldsymbol{x}^{(h)}\right)\right)\left(\prod_{h=1}^{H+1} q\left(\boldsymbol{z}^{(h)}\right)\right)$$
$$\left(\prod_{i=L+1}^{N}\prod_{h=1}^{H+1} p\left(f_i^{(h)}|\boldsymbol{z}^{(h)},\hat{\boldsymbol{x}}_i^{(h)}\right)\right), \tag{22}$$

where $q(\boldsymbol{\tau}), q\left(\boldsymbol{x}^{(h)}\right)$ and $q\left(\boldsymbol{z}^{(h)}\right)$ are respectively the variational posterior distributions related to the precisions $\boldsymbol{\tau}$, the latent variables $\boldsymbol{x}^{(h)}$ and the inducing points $\boldsymbol{z}^{(h)}$. The distribution $p\left(f_i^{(h)}|\boldsymbol{z}^{(h)},\hat{\boldsymbol{x}}_i^{(h)}\right)$ can be found by using the Gaussian conditioning property:

$$p\left(f_i^{(h)}|\boldsymbol{z}^{(h)},\hat{\boldsymbol{x}}_i^{(h)}\right) = \mathcal{N}\left(f_i^{(h)}|\left[\boldsymbol{a}_f^{(h)}\right]_i,\left[\boldsymbol{\Sigma}_f^{(h)}\right]_{ii}\right), \tag{23}$$

where

$$\boldsymbol{a}_f^{(h)} = \boldsymbol{K}_{fz}^{(h)}\left(\boldsymbol{K}_z^{(h)}\right)^{-1}\boldsymbol{z}^{(h)} \text{and} \boldsymbol{\Sigma}_f^{(h)} = \boldsymbol{K}_f^{(h)} - \boldsymbol{K}_{fz}^{(h)}\left(\boldsymbol{K}_z^{(h)}\right)^{-1}\left(\boldsymbol{K}_{fz}^{(h)}\right)^{\top},$$

given that the zero-mean joint Gaussian distribution below holds for all layers:

$$p\left(\boldsymbol{f}^{(h)},\boldsymbol{z}^{(h)}\right) = \mathcal{N}\left(\begin{bmatrix}\boldsymbol{f}^{(h)}\\ \boldsymbol{z}^{(h)}\end{bmatrix}|\boldsymbol{0},\begin{bmatrix}\boldsymbol{K}_f^{(h)} & \boldsymbol{K}_{fz}^{(h)}\\ \left(\boldsymbol{K}_{fz}^{(h)}\right)^{\top} & \boldsymbol{K}_z^{(h)}\end{bmatrix}\right), \tag{24}$$

In the above expressions, $\boldsymbol{K}_f^{(h)} \in \mathbb{R}^{(N-L)\times(N-L)}$ is the standard kernel matrix obtained from $\hat{\boldsymbol{x}}^{(h)}$, $\boldsymbol{K}_z^{(h)} \in \mathbb{R}^{M\times M}$ is the sparse kernel matrix calculated from the pseudo-inputs $\boldsymbol{\zeta}_j^{(h)}|_{j=1}^{M}|$ and $\left[\boldsymbol{K}_{fz}^{(h)}\right]_{ij} = k\left(\hat{\boldsymbol{x}}_i^{(h)},\boldsymbol{\zeta}_j^{(h)}\right) \in \mathbb{R}^{(N-L)\times M}$. The subindex $[\cdot]_i$ indicates the ith element of the vector $\boldsymbol{a}_f^{(h)}$, while the double subindex $[\cdot]_{ii}$ indicates the ith element of the diagonal of the matrix $\boldsymbol{\Sigma}_f^{(h)}$. It should be noted that we do not need to compute the full covariance matrix $\boldsymbol{K}_f^{(h)}$, but only its N diagonal elements, a feature inherited from the sparse approximation. We also recall that the number of inducing points M is usually much lower than the number of samples N, i.e., $M \ll N$.

In order to make the integrals in Eq. (21) tractable, we continue to follow the standard variational framework [55] and consider a factorized mean-field approximation for $q(\boldsymbol{\tau})$ and $q\left(\boldsymbol{x}^{(h)}\right)$, which are given by

$$q(\boldsymbol{\tau}) = \prod_{i=L+1}^{N}\Gamma(\tau_i|a_i,b_i),$$

where a_i, b_i, $\mu_i^{(h)}$ and $\lambda_i^{(h)}$ are variational parameters that characterize the posterior approximations. It should be noted that, while the hyperparameters of the gamma prior α and β are global, each sample is related to its own set of variational posterior parameters a_i and b_i.

Following [44], it turns out that the variational distribution $q\left(\boldsymbol{z}^{(h)}\right)$ that maximizes the bound is a multivariate Gaussian, which can be generically written as $q\left(\boldsymbol{z}^{(h)}\right) = \mathcal{N}\left(\boldsymbol{z}^{(h)}|\boldsymbol{m}^{(h)},\boldsymbol{\Sigma}^{(h)}\right)$, where $\boldsymbol{m}^{(h)}$ and $\boldsymbol{\Sigma}^{(h)}$ are additional variational parameters respectively related to the mean and covariance matrix of the distribution.

[8] $\int_{\boldsymbol{\tau},\boldsymbol{f},\boldsymbol{x},\boldsymbol{z}}$ is equivalent to $\int \mathrm{d}\boldsymbol{\tau}\mathrm{d}\boldsymbol{x}^{(1)}\cdots\mathrm{d}\boldsymbol{x}^{(H)}\mathrm{d}\boldsymbol{f}^{(1)}\cdots\mathrm{d}\boldsymbol{f}^{(H+1)}\mathrm{d}\boldsymbol{z}^{(1)}\cdots\mathrm{d}\boldsymbol{z}^{(H+1)}$ and covers all the possible values taken by the integrated variables.

Fortunately, those can be found analytically and optimally eliminated from the expressions, as described in [44,45].

Inserting the factorized variational distribution Q back into Eq. (21) we are able to cancel out the intractable terms $p\left(f_i^{(h)}|\boldsymbol{z}^{(h)}, \hat{\boldsymbol{x}}_i^{(h)}\right)$ inside the logarithm. After solving the tractable integrals and some algebraic manipulation, we finally get the final form of the REVARB-t lower bound:

$$
\begin{aligned}
&\log p(\boldsymbol{y}) \geq \\
&-\frac{N-L}{2}\sum_{h=1}^{H}\log 2\pi\sigma_h^2 - \frac{N-L}{2}\log 2\pi + \frac{1}{2}\sum_{i=L+1}^{N}\left(\psi(a_i) - \log b_i\right) \\
&-\frac{1}{2}\left(\boldsymbol{y}^\top\boldsymbol{R}\boldsymbol{y} + \Psi_0'^{(H+1)} - \mathrm{Tr}\left(\left(\boldsymbol{K}_z^{(H+1)}\right)^{-1}\boldsymbol{\Psi}_2'^{(H+1)}\right)\right) \\
&+\frac{1}{2}\log\left|\boldsymbol{K}_z^{(H+1)}\right| - \frac{1}{2}\log\left|\boldsymbol{K}_z^{(H+1)} + \boldsymbol{\Psi}_2'^{(H+1)}\right| \\
&+\frac{1}{2}\boldsymbol{y}^\top\boldsymbol{\Psi}_1'^{(H+1)}\left(\boldsymbol{K}_z^{(H+1)} + \boldsymbol{\Psi}_2'^{(H+1)}\right)^{-1}\left(\boldsymbol{\Psi}_1'^{(H+1)}\right)^\top\boldsymbol{y} - \mathrm{KL}(q(\boldsymbol{\tau})||p(\boldsymbol{\tau})) \\
&+\sum_{h=1}^{H}\left\{-\frac{1}{2\sigma_h^2}\left(\sum_{i=L+1}^{N}\lambda_i^{(h)} + \left(\boldsymbol{\mu}^{(h)}\right)^\top\boldsymbol{\mu}^{(h)} - \mathrm{Tr}\left(\left(\boldsymbol{K}_z^{(h)}\right)^{-1}\boldsymbol{\Psi}_2^{(h)}\right)\right)\right. \\
&-\frac{1}{2\sigma_h^2}\Psi_0^{(h)} + \frac{1}{2}\log\left|\boldsymbol{K}_z^{(h)}\right| - \frac{1}{2}\log\left|\boldsymbol{K}_z^{(h)} + \frac{1}{\sigma_h^2}\boldsymbol{\Psi}_2^{(h)}\right| \\
&+\frac{1}{2(\sigma_h^2)^2}\left(\boldsymbol{\mu}^{(h)}\right)^\top\boldsymbol{\Psi}_1^{(h)}\left(\boldsymbol{K}_z^{(h)} + \frac{1}{\sigma_h^2}\boldsymbol{\Psi}_2^{(h)}\right)^{-1}\left(\boldsymbol{\Psi}_1^{(h)}\right)^\top\boldsymbol{\mu}^{(h)} \\
&\left.-\sum_{i=1}^{N}\int_{x_i^{(h)}} q\left(x_i^{(h)}\right)\log q\left(x_i^{(h)}\right) + \sum_{i=1}^{L}\int_{x_i^{(h)}} q\left(x_i^{(h)}\right)\log p\left(x_i^{(h)}\right)\right\},
\end{aligned}
\tag{27}
$$

where we have defined the statistics $\Psi_0^{(h)} = \mathrm{Tr}\left(\langle\boldsymbol{K}_f^{(h)}\rangle_{q(\boldsymbol{x})}\right)$, $\boldsymbol{\Psi}_1^{(h)} = \langle\boldsymbol{K}_{fz}^{(h)}\rangle_{q(\boldsymbol{x})}$ and $\boldsymbol{\Psi}_2^{(h)} = \langle\left(\boldsymbol{K}_{fz}^{(h)}\right)^\top\boldsymbol{K}_{fz}^{(h)}\rangle_{q(\boldsymbol{x})}$ for the hidden layers, i.e., $1 \leq h \leq H$. Due to the robust RGP-t output layer, we have slightly different statistics for $h = H+1$:

$$
\Psi_0'^{(H+1)} = \mathrm{Tr}\left(\boldsymbol{R}\langle\boldsymbol{K}_f^{(H+1)}\rangle_{q(\boldsymbol{x})}\right),
$$

where $\boldsymbol{R} = \mathrm{diag}\left(\frac{a_{L+1}}{b_{L+1}}, \ldots, \frac{a_N}{b_N}\right)$.

In the previous expressions, $\mathrm{diag}(\cdot)$ builds a diagonal matrix from its arguments, $\mathrm{Tr}(\cdot)$ denotes the trace of a matrix, and $\langle\cdot\rangle_{q(\boldsymbol{x})}$ denotes expectation with respect to the variational distribution $q(\boldsymbol{x})$ of the latent dynamical variables, where the layer indexes were omitted. When the exponentiated quadratic kernel in Eq. (2) is chosen, all the expectations can be solved in closed form, as detailed in [45]. The lower bound includes some entropy integrals at the end of Eq. (27), which involve Gaussians and are, hence, also tractable.

The REVARB-t bound in Eq. (27) also contains the Kullback–Leibler divergence $\mathrm{KL}(q(\boldsymbol{\tau})||p(\boldsymbol{\tau}))$ between two factorized gamma distributions, which is given by

$$
\begin{aligned}
\mathrm{KL}(q(\boldsymbol{\tau})||p(\boldsymbol{\tau})) \;=\; &\sum_{i=L+1}^{N}\left\{(a_i-\alpha)\psi(a_i) - \log\Gamma(a_i) + \log\Gamma(\alpha)\right. \\
&\left.+\alpha(\log b_i - \log\beta) + a_i\frac{\beta - b_i}{b_i}\right\},
\end{aligned}
\tag{31}
$$

where $\psi(\cdot)$ is the digamma distribution, i.e., the first derivative of the gamma function.

Remark 4. The second and fifth lines in Eq. (27) show the only terms that include the observations $\boldsymbol{y}$. In those terms, introduced by the proposed REVARB-t framework, the value of each output y_i is always weighted by the ratio a_i/b_i in the diagonal of the matrix $\boldsymbol{R}$, which comes from the expectation of a gamma distribution. For observations containing outliers, the value of this ratio is much lower than regular observations, reducing their influence in the bound. Furthermore, the inspection of the optimized variational parameters a_i and b_i *per se* can be used as a method to detect outliers in the estimation data.

Remark 5. The REVARB-t model selection step is performed by jointly optimizing the kernel hyperparameters and the variational parameters, including the pseudo-inputs $\boldsymbol{\zeta}_j^{(h)}$ and the gamma posterior parameters a_i and b_i, by maximizing the lower bound on the log-marginal likelihood. Any gradient-based optimization algorithm can be used, following the analytical gradients of Eq. (27). In this work we use the well known Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm available in the *R* implementation of the standard *stats* package [56]. Alternatively, instead of manually taking the gradients of the lower bound in Eq. (27), one could use automatic differentiation tools to ease the implementation, such as the ones available in the Theano [57] or TensorFlow [58] programming frameworks.

4.1. Free simulation with approximate uncertainty propagation

We are interested in using the proposed RGP-t model to perform free simulation (i.e., infinite horizon prediction). Thus, given a sequence of new control inputs, past inputs and past predictions are iteratively inserted back into the model's regressors in order to infer the next outputs. It should be noted that, each new prediction consists of a fully characterized distribution which takes into account the uncertainty of past predictions.

Since the input of each layer in a given iteration is a distribution that models the uncertainty in that moment, the computation of the exact prediction is intractable, in opposition to standard GP regression. However, we can still calculate its moments and approximate it by a Gaussian for further predictions, following the methodology presented in [14] for multi-step ahead GP predictions. The result is a set of modified predictive equations from the original RGP model, presented in [52]. Given a new regressor vectors $\hat{\boldsymbol{x}}_*^{(h)}$, the mean $\mu_*^{(h)}$ and variance $\lambda_*^{(h)}$ of the predictions $f_*^{(h)}$ within each layer of the RGP-t model are calculated by approximately propagating uncertainty between each layer, using the equations below:

$$
\mu_*^{(h)} = \left(\boldsymbol{B}^{(h)}\right)^\top\left(\boldsymbol{\Psi}_{1*}^{(h)}\right)^\top,
$$

where $\hat{\boldsymbol{x}}_*^{(h)}$ is defined similarly to Eq. (19), $\boldsymbol{B}^{(h)} = \sigma_h^{-2}\left(\boldsymbol{K}_z^{(h)} + \sigma_h^{-2}\boldsymbol{\Psi}_2^{(h)}\right)^{-1}\left(\boldsymbol{\Psi}_1^{(h)}\right)^\top\boldsymbol{\mu}^{(h)}$, for $1 \leq h \leq H$, and $\boldsymbol{B}'^{(H+1)} = \left(\boldsymbol{K}_z^{(H+1)} + \boldsymbol{\Psi}_2'^{(H+1)}\right)^{-1}\left(\boldsymbol{\Psi}_1'^{(H+1)}\right)^\top\boldsymbol{y}$.

In the expression for $\boldsymbol{B}'^{(H+1)}$, the training outputs $\boldsymbol{y}$ are once again weighted by the diagonal of the matrix $\boldsymbol{R}$, inside $\boldsymbol{\Psi}_1'^{(H+1)}$, which reduces the influence of the outliers in the predictions. The terms $\Psi_{0*}^{(h)}$, $\boldsymbol{\Psi}_{1*}^{(h)}$ and $\boldsymbol{\Psi}_{2*}^{(h)}$ are computed as before (see Eqs. (28)–(30)), but instead of the distributions $q\left(x_i^{(h)}\right)$ we use the new Gaussian approximation $q\left(x_*^{(h)}\right) = \mathcal{N}\left(x_*^{(h)}|\mu_*^{(h)}, \lambda_*^{(h)}\right)$ and replace $\boldsymbol{K}_f^{(h)}$ and $\boldsymbol{K}_{fz}^{(h)}$ respectively with $k_{**}^{(h)} = k\left(\hat{\boldsymbol{x}}_*^{(h)}, \hat{\boldsymbol{x}}_*^{(h)}\right) \in \mathbb{R}$ and $\boldsymbol{k}_{*z}^{(h)} = \left[k\left(\hat{\boldsymbol{x}}_*^{(h)}, \boldsymbol{\zeta}_1^{(h)}\right), \ldots, k\left(\hat{\boldsymbol{x}}_*^{(h)}, \boldsymbol{\zeta}_M^{(h)}\right)\right] \in \mathbb{R}^M$.

As a final remark, we emphasize that the REVARB-t predictive equations are consistent with our choice to evaluate the models by free simulation, since we do not use the past actually observed

Table 2
Details of the six artificial datasets used in the computational experiments. U(A, B) is a random number uniformly distributed between A and B.

#	Output	Input/samples		Noise
		Estimation	Test	
1	$y_i = \frac{y_{i-1}y_{i-2}(y_{i-1}+2.5)}{1+y_{i-1}^2+y_{i-2}^2} + u_{i-1}$	$u_i = \mathrm{U}(-2, 2)$ 300 samples	$u_i = \sin(2\pi i/25)$ 100 samples	$\mathcal{N}(0, 0.29)$
2	$y_i = \frac{y_{i-1}}{1+y_{i-1}^2} + u_{i-1}^3$	$u_i = \mathrm{U}(-2, 2)$ 300 samples	$u_i = \sin(2\pi i/25) + \sin(2\pi i/10)$ 100 samples	$\mathcal{N}(0, 0.65)$
3	$y_i = 0.8y_{i-1} + (u_{i-1} - 0.8)u_{i-1}(u_{i-1} + 0.5)$	$u_i = \mathrm{U}(-1, 1)$ 300 samples	$u_i = \sin(2\pi i/25)$ 100 samples	$\mathcal{N}(0, 0.07)$
4	$y_i = 0.3y_{i-1} + 0.6y_{i-2} + 0.3\sin(3\pi u_{i-1}) + 0.1\sin(5\pi u_{i-1})$	$u_i = \mathrm{U}(-1, 1)$ 500 samples	$u_i = \sin(2\pi i/250)$ 500 samples	$\mathcal{N}(0, 0.18)$
5	$y_i = y_{i-1} - 0.5\tanh(y_{i-1} + u_{i-1}^3)$	$u_i = \mathcal{N}(u_i\|0, 1)$ $-1 \le u_i \le 1$ 150 samples	$u_i = \mathcal{N}(u_i\|0, 1)$ $-1 \le u_i \le 1$ 150 samples	$\mathcal{N}(0, 0.0025)$
6	$x_i^{(1)} = x_{i-1}^{(2)}$ $x_i^{(2)} = -\frac{3}{16}\frac{x_{i-1}^{(1)}}{\left(1+\left(x_{i-1}^{(2)}\right)^2\right)} + x_{i-1}^{(2)} + u_{i-1}$ $y_i = x_i^{(1)}$	$u_i = \mathrm{U}(-0.5, 0.5)$ 300 samples	$u_i = \mathrm{U}(-0.5, 0.5)$ 300 samples	$\mathcal{N}(0, 0.0065)$

test values, but rather the predicted values. Moreover, such approach considers the simulation uncertainty by using the whole distribution $q\left(\boldsymbol{x}_*^{(h)}\right)$ of past latent variables to predict the next step, instead of just the mean value.

In Algorithm 2 we summarize all the steps regarding the use of the REVARB-t framework in robust system identification. Note that at the end of the estimation step it is possible to check which estimation samples probably contain outliers. Besides, each output during free simulation on test data is approximated by a Gaussian distribution.

Algorithm 2. RGP-t for system identification with REVARB-t.

- Estimation step
Require: $\boldsymbol{u} \in \mathbb{R}^N$ (control inputs), $\boldsymbol{y} \in \mathbb{R}^N$ (outputs), H (number of hidden layers), M (number of inducing points), L (latent order lag), L_u (input order lag)
Initialize kernel hyperparameters and variational parameters;
repeat
Compute the evidence lower bound with Eq. (27);
Compute the analytical gradients of Eq. (27) with respect to the unknown parameters;
Update parameters with a gradient-based method (e.g. BFGS);
until convergence or maximum number or iterations
Output the optimized parameters;
Check the ratios $a_i/b_i|_{i=L+1}^N$, the smallest values are related to the estimation samples which probably contain outliers;
- Free simulation with test data
Require: Test control inputs $\boldsymbol{u}_* \in \mathbb{R}^{N_*}$ and the previously estimated RGP-t model
for $i = 1 : N_*$ **do**
for $h = 1 : H$ **do**
Compute the predictive mean $\mu_{*i}^{(h)}$ and variance $\lambda_{*i}^{(h)}$ with Eqs. (32) and (34);
Update the variational distributions (Eq. (26)) with $q\left(x_{*i}^{(h)}\right) = \mathcal{N}\left(\mu_{*i}^{(h)}, \lambda_{*i}^{(h)}\right)$;
end for
Compute the predictive mean $\mu_{*i}^{(H+1)}$ and variance $\lambda_{*i}^{(H+1)}$ of the output layer with Eqs. (33) and (35);
Output $f_{*i}^{(H+1)} \sim \mathcal{N}\left(\mu_{*i}^{(H+1)}, \lambda_{*i}^{(H+1)}\right)$;
end for

5. Computer experiments and discussion

In this section we perform several computational experiments in order to evaluate the proposed RGP-t and compare it with the standard GP-NARX and the robust GP-tVB and GP-RLARX models. The experiments involve six artificial benchmarks available in the system identification literature and two datasets related to process industry systems, available in the DaISy (**Da**tabase for the **I**dentification of **Sy**stems) repository [59]. In the latter two cases we also include experiments with the standard RGP model with Gaussian likelihood.

We have also made available a simple automatic working example in the repository accessible via the link https://github.com/cesarlincoln/jpc_rgpt.

5.1. Artificial benchmarks

We complement the experiments performed in [48] by applying the newly introduced RGP-t model to the same five artificial datasets and scenarios used in that work and including a sixth dataset. All the artificial benchmarks are summarized in Table 2. The first four were introduced in the seminal work of Narendra and Parthasarathy [60]. The fifth was generated as in [12], while the sixth dataset is derived from the artificial plant described in [61]. The outliers were generated by sampling from $\sigma(\boldsymbol{y}) \times \mathcal{T}(0, 2)$, i.e., a Student-$t$ distribution with zero mean and 2 degrees of freedom scaled by the standard deviation of the original estimation data. We considered scenarios without outliers and with 5%, 10%, 15%, 20%, 25% and 30% of the estimation samples uniformly chosen to be contaminated by outliers. In all experiments we used a RGP-t model with 1 hidden (transition) layer and 1 output (observation) layer. The number of inducing points M was fixed to 10% of the number of samples and the pseudo-inputs were initialized using the PAM (Partition Around Medoids [62])[9] algorithm on the estimation regressors. More specifically, we used the deterministic R implementation of the *cluster* package [63].

The obtained root mean square errors (RMSE) are presented in the line charts of Fig. 2. The robustness of the GP-RLARX model was already praised in [48] and its latent autoregressive structure was able to pair up with the heavy tailed Student-t likelihood and be more resilient to outliers than GP-tVB. The RGP-t model maintains those features and incorporates additional representational capabilities to the dynamical modeling. Thus, besides the dataset Artificial 1, where RGP-t performed quite similar to GP-RLARX, in

[9] PAM is an algorithm for the k-medoids clustering method, related to the more common K-means algorithm.

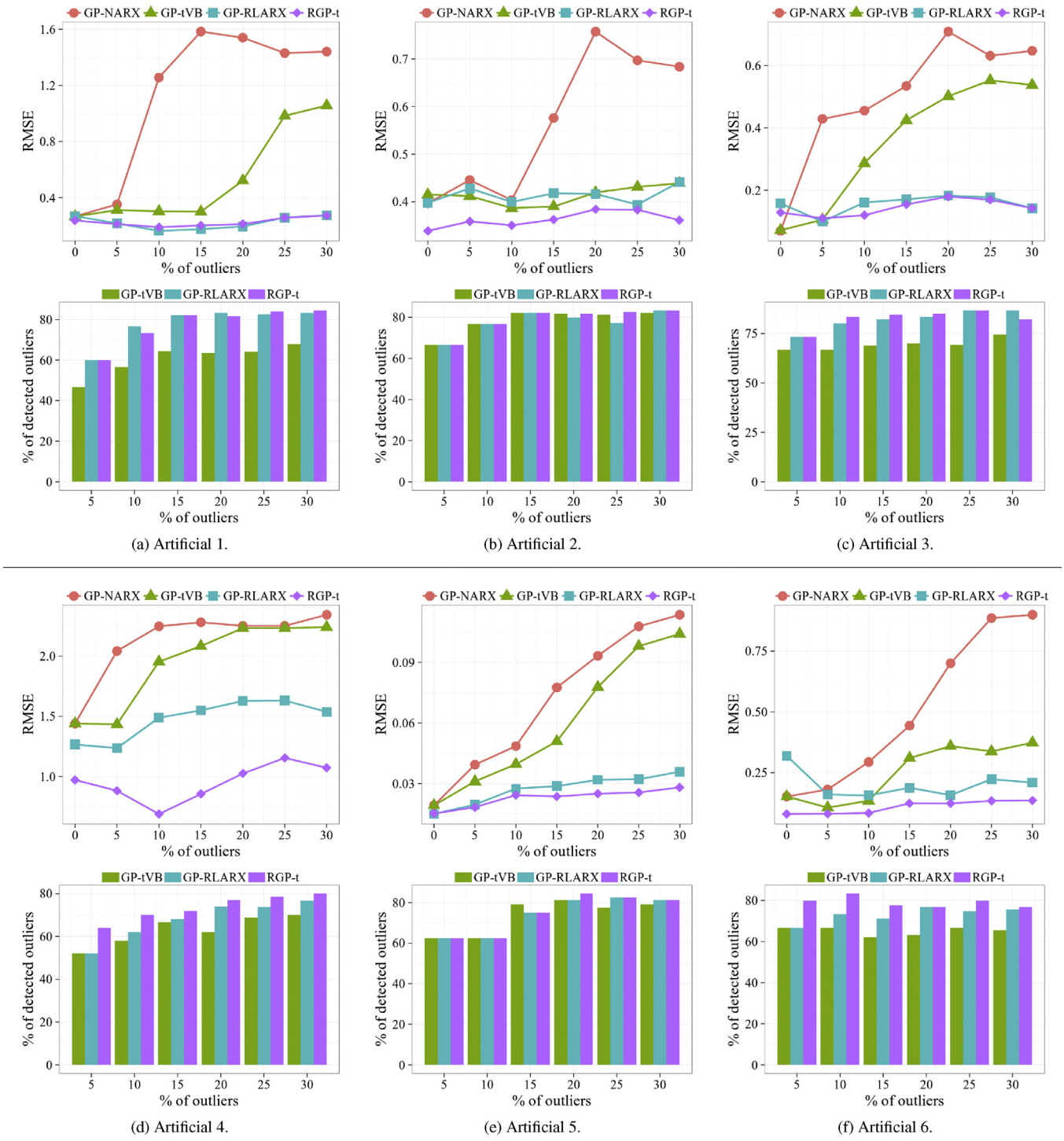


Fig. 2. Line charts for the RMSE values related to the free simulation on test data with different levels of contamination by outliers. The bar plots indicate the percentage of outliers detected by the robust models using the variational framework.

almost all the other scenarios the multilayered structure of RGP-*t* achieved better results. For instance, in the case of Artificial 2, 4 and 6 datasets, large gain in performance occurred in all levels of contamination, which highlights the powerful hierarchical structure of the RGP-*t* model.

As mentioned in Section 4, the variational framework of the three robust models, i.e., GP-tVB, GP-RLARX and RGP-*t*, allows for a practical way of detecting estimation samples that contain outliers. After the model optimization step, by computing the ratios a_i/b_i, $i = 1, \ldots, N$, along the estimation data and sorting the values, we can select the smallest ones, which correspond to smallest variational precisions (or largest variances), and associate them with impulsive noise samples. The bar plots in Fig. 2 show the results of the application of such methodology, where we have selected in each case the expected amount of outliers (5% to 30%).

The three models are able to detect most of the outliers in the corrupted scenarios. Interestingly, although all the models use the same Student-*t* likelihood, in many cases the GP-RLARX and RGP-*t* were able to detect more outliers than GP-tVB, with RGP-*t* being slightly better overall, mainly for Artificial 4 and 6 datasets.

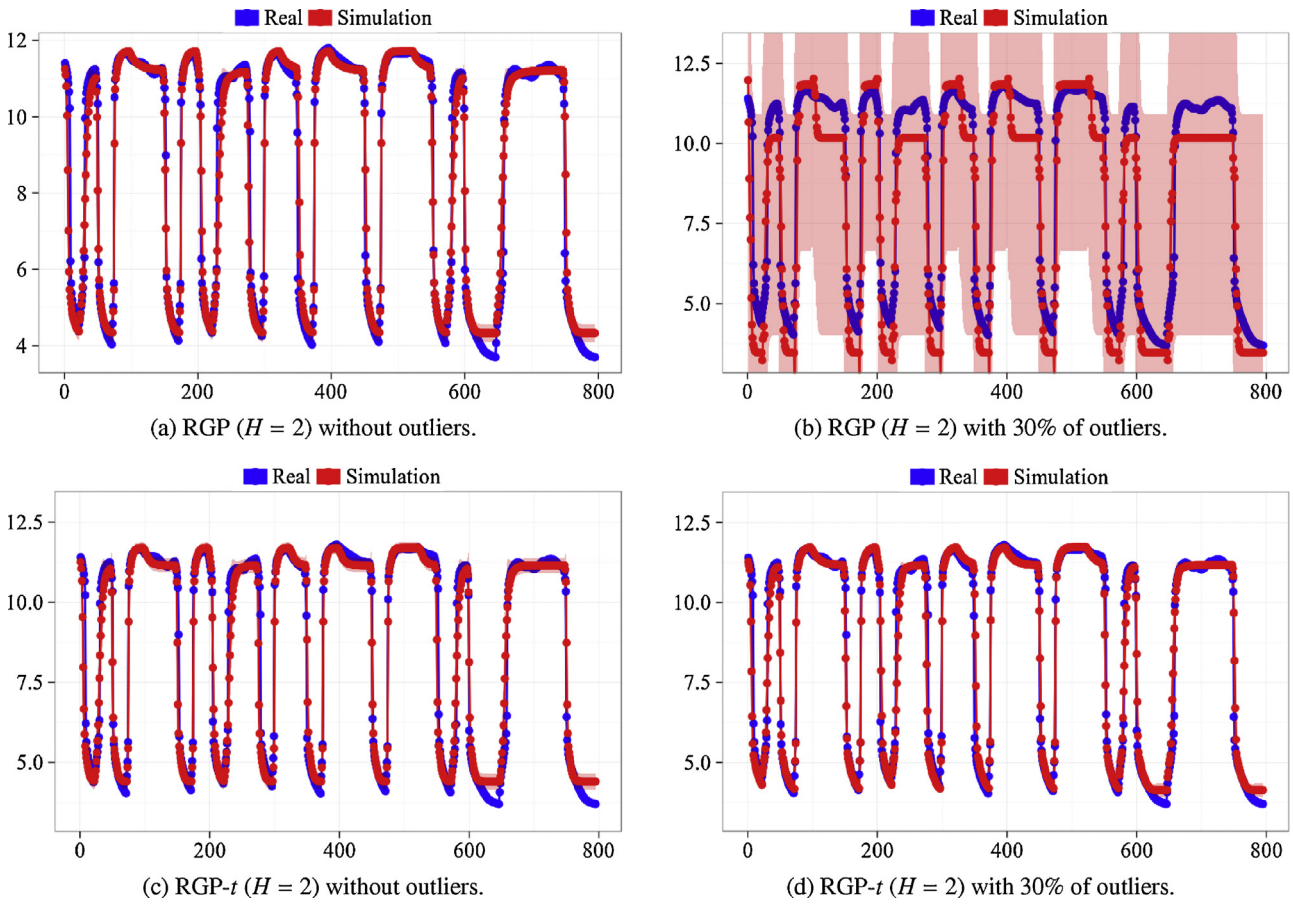


Fig. 3. Free simulation on test data after estimation on the pH train dataset without and with outliers.

It should be noted that, in some scenarios, such as the ones of the Artificial 5 dataset, although GP-tVB has been able to detect almost the same amount of outliers of the other robust models, its performance in terms of test prediction was considerably worse than GP-RLARX and specially RGP-t. Such observation once again stresses the importance of the latent autoregressive structure in the robust system identification task.

5.2. pH data

We now evaluate the GP models with the pH dataset available in the DaISy repository [59]. The data comes from a pH neutralization process in a constant volume stirring tank. The control input is the base solution flow and the output is the pH value of the solution in the tank. We apply the first 200 samples for estimation and the next 800 samples for model validation. Two scenarios were considered, one without outliers and other where the estimation data was contaminated with 30% of outliers, sampled from $\sigma(\boldsymbol{y}) \times \mathcal{T}(0, 2)$, where $\sigma(\boldsymbol{y})$ is the standard deviation of the original estimation data. No Gaussian noise was added to the data. We chose the orders $L = L_u = 5$ for the regressors. Besides the models used in Section 5.1, we also evaluate the original RGP model with Gaussian likelihood (with 1 and 2 hidden layers).

In Table 3 we report the test RMSE values and the average NLPD (negative log predictive density), given by $\text{NLPD} = \frac{1}{2}\log 2\pi + \frac{1}{2N}\sum_{i=1}^{N}\left[\log\hat{\sigma}_i^2 + \frac{(y_i-\hat{\mu}_i)^2}{\hat{\sigma}_i^2}\right]$, where $\hat{\mu}_i$ is the ith predicted mean and $\hat{\sigma}_i^2$ is the corresponding predicted variance. The NLPD is a metric of the type "the lower, the better", since it indicates a favorable balance between the squared error and the compactness of the uncertain prediction. When applicable, the amount of outliers correctly detected by the robust models is also shown.

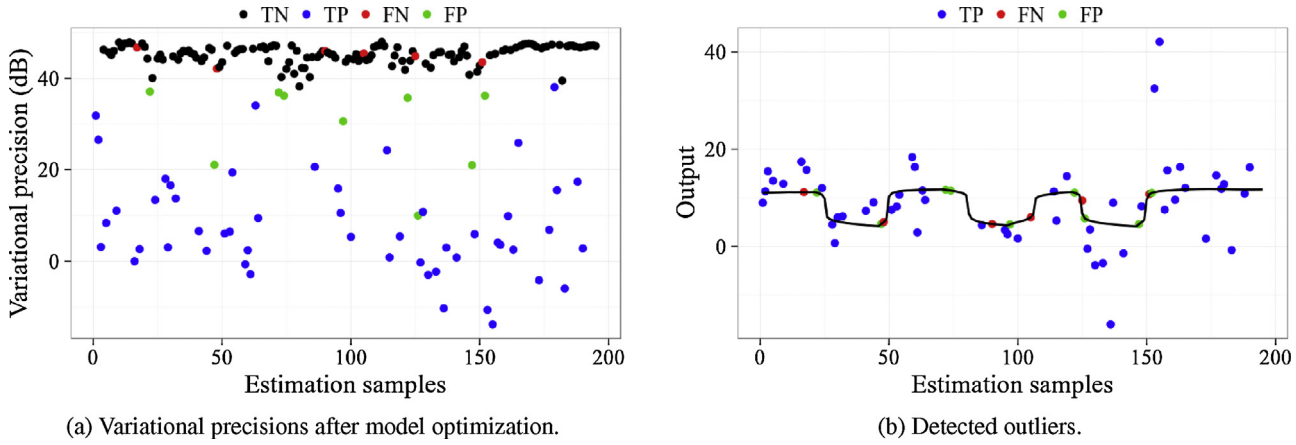
Table 3
Results for free simulation on test data after estimation on the pH dataset without and with outliers.

	Without outliers		With 30% of outliers		
	RMSE	NLPD	RMSE	NLPD	% detected
GP-NARX	0.8716	1.8435	1.2735	2.3344	–
GP-tVB	0.9834	4264.0	1.0619	28.6278	76.7
GP-RLARX	0.8305	1.9181	0.7776	1.4777	75.0
RGP ($H = 1$)	0.6658	1.7751	0.9299	1.8496	–
RGP ($H = 2$)	**0.6616**	1.5625	1.2258	2.1177	–
RGP-t ($H = 1$)	0.6661	**0.6363**	0.6221	1.4663	81.7
RGP-t ($H = 2$)	0.6796	1.4283	**0.6192**	**1.1978**	**85.0**

Best values in bold.

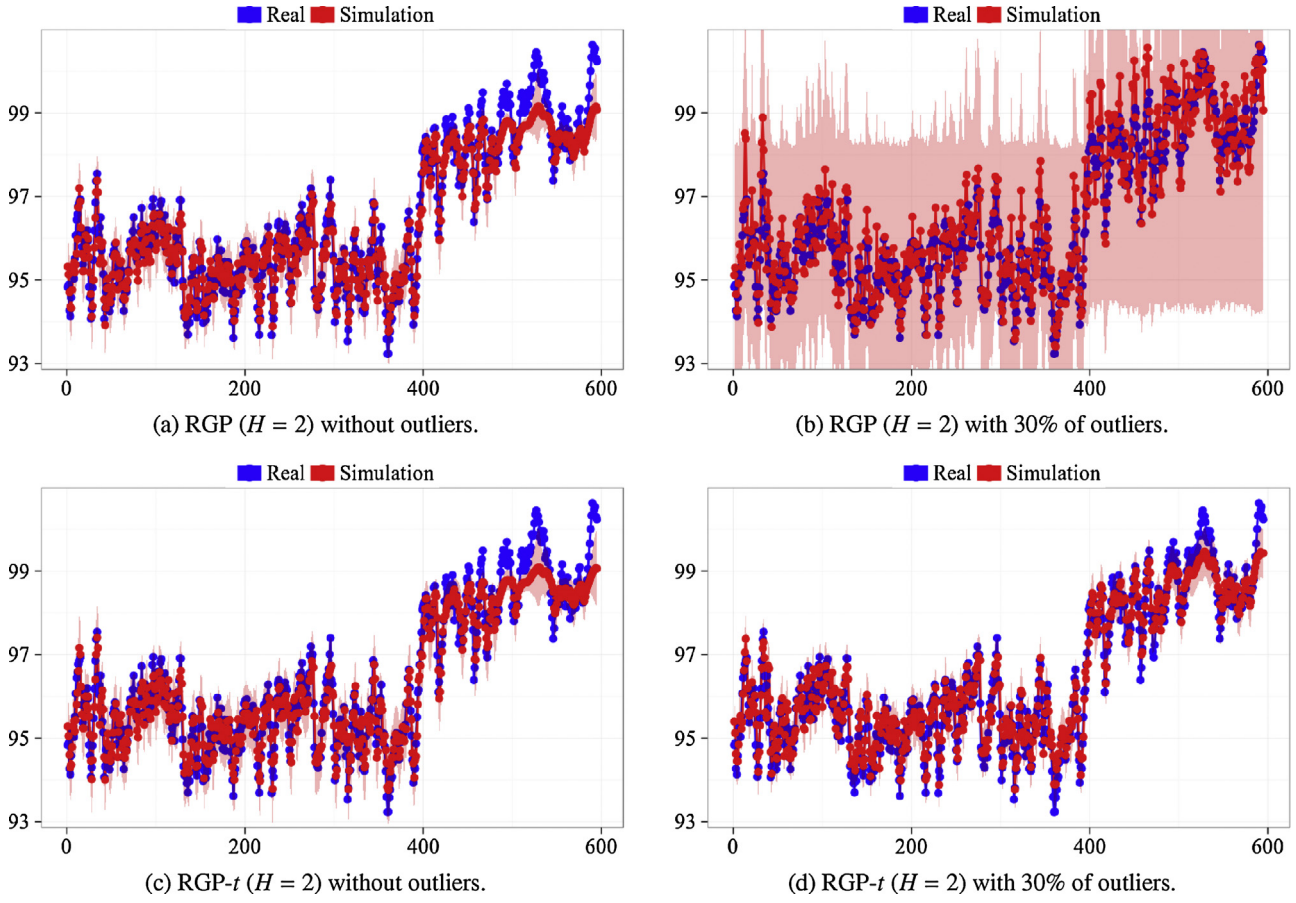
The corrupted scenario is quite extreme, with few estimation samples and a large contamination rate, but both RGP-t models are still able to reasonably learn the system dynamics. Actually, the RMSE values obtained for those models were even lower in the corrupted case than in the non-corrupted case, which can be explained by the fact that the former follows the expected Student-t noise prior of the robust models. Similar behavior was also observed in some of the results presented in Fig. 2 in Section 5.1.

The free simulation outputs on test data for the best model, the RGP-t with 2 hidden layers, and its non-robust counterpart, RGP with 2 hidden layers, are presented in Fig. 3, where the shaded area indicates ± 2 standard deviations around the predicted mean $\hat{\mu}_i$. The RGP-t model seems "overconfident", since its shaded area is barely visible, but it still has presented superior performance for all three considered metrics in the corrupted scenario. Note that



(a) Variational precisions after model optimization. (b) Detected outliers.

Fig. 4. Outlier detection by the RGP-*t* model with 2 hidden layers for the pH estimation data in the scenario with 30% of outliers. TN are the true negatives (correctly classified as non-outliers), TP are the true positives (correctly classified as outliers), FN are the false negatives (undetected outliers) and FP are the false positives (regular samples misclassified as outliers). In the right side, the black line indicates the original training output.



(a) RGP ($H = 2$) without outliers. (b) RGP ($H = 2$) with 30% of outliers.

(c) RGP-*t* ($H = 2$) without outliers. (d) RGP-*t* ($H = 2$) with 30% of outliers.

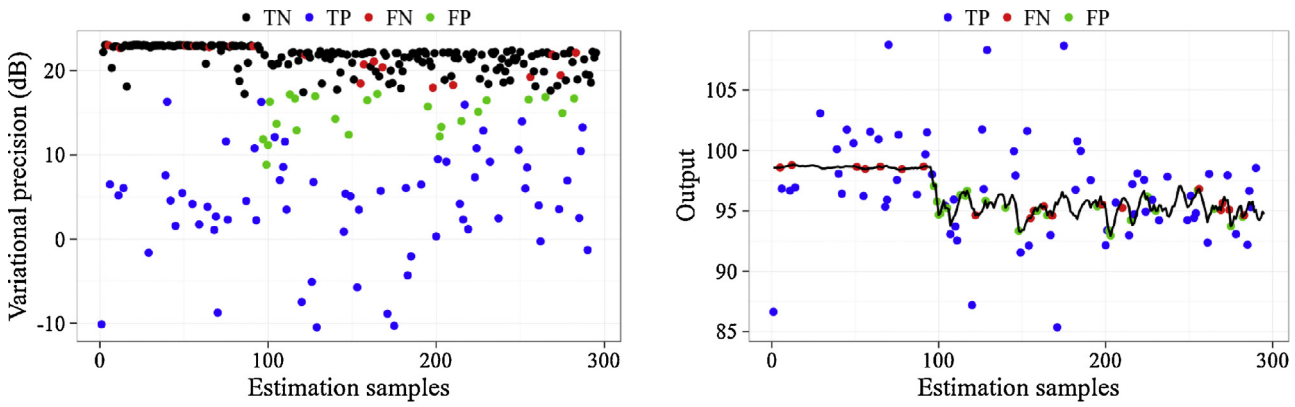
Fig. 5. Free simulation on test data after estimation on the Heat Exchanger train dataset without and with outliers.

the output of the non-robust RGP model, although much worse in the case with outliers, is not so "wild" due to the presence of the control input, which is not contaminated.

In Fig. 4 we report results on the outlier detection capability of the proposed RGP-*t* model. On the left side the variational precisions, i.e., the ratios a_i/b_i, $i = L+1, \ldots, N$, are shown, where we can see a clear difference in the magnitude of the optimized precisions between outliers and regular samples, the black dots correctly identified as non-outliers (true negatives). On the right side we confirm that all the severe outliers were correctly detected (true positives, represented by the blue dots), since the undetected ones (false negatives, red dots), are actually not far away from the original data (black line). The points misclassified as outliers (false positives, green dots), whose amount was not enough to compromise the performance of the model, are also shown.

5.3. Heat Exchanger data

We conclude the experiments with the Heat Exchanger dataset, also available in the DaISy repository [59]. The data comes from



(a) Variational precisions after model optimization. (b) Detected outliers.

Fig. 6. Outlier detection by the RGP-*t* model with 2 hidden layers for the Heat Exchanger estimation data in the scenario with 30% of outliers. TN are the true negatives (correctly classified as non-outliers), TP are the true positives (correctly classified as outliers), FN are the false negatives (undetected outliers) and FP are the false positives (regular samples misclassified as outliers). In the right side, the black line indicates the original training output.

Table 4
Results for free simulation on test data after estimation on the Heat Exchanger dataset without and with outliers.

	Without outliers		With 30% of outliers		
	RMSE	NLPD	RMSE	NLPD	% detected
GP-NARX	0.4816	0.8360	1.7269	2.0772	–
GP-tVB	0.7030	13.4332	0.5627	1.7514	**77.8**
GP-RLARX	0.5532	0.8979	0.4362	0.7374	76.7
RGP ($H=1$)	**0.4223**	**0.6893**	0.7885	1.5402	–
RGP ($H=2$)	0.4638	2.2295	0.5977	1.5324	–
RGP-*t* ($H=1$)	0.4745	1.6603	0.4233	0.8927	74.4
RGP-*t* ($H=2$)	0.4563	1.4855	**0.4087**	**0.7039**	74.4

Best values in bold.

a liquid-saturated steam heat exchanger, where water is heated by pressurized saturated steam through a copper tube. The output variable is the outlet liquid temperature and the input variable is the liquid flow rate. As in the experiment with the pH data, we evaluate the GP models in a scenario without outliers and another with 30% of outliers, which were sampled from the scaled distribution $\sigma(\boldsymbol{y}) \times \mathcal{T}(0, 2)$. In both scenarios Gaussian noise was not added to the data. The estimation set contains 300 samples (starting from the 101-th sample, since the first 100 are equal), while the test set contains the next 600 samples. We fixed the orders $L = L_u = 5$ for the regressors.

Numerical results are shown in Table 4, where the robustness of both RGP-*t* models is again verified, with the 2-layered variant presenting the best RMSE and NLPD values. The outputs of the free simulation test for the latter model and its non-robust RGP counterpart are presented in Fig. 5, where we can observe that the RGP-*t* is unaffected by the outliers and actually presents the best performance. One more time the simulation of the non-robust RGP model in the contaminated scenario, although worse than before the outliers inclusion, is not completely far off the real output due to the presence of the uncontaminated exogenous control input.

In Fig. 6 we show the outliers detected by the RGP-*t* model. Even though this time the model did not present the best detection rate among the robust models, it is noticeable that most of the more relevant outliers were found. We emphasize that the low variational precision values (blue dots) shown in the left side are applied as weights to the correspondent output estimation samples, which greatly hinders any spurious effect caused by outliers on the model learning and predictive capabilities.

6. Conclusions

In the present paper we tackled the task of nonlinear system identification in the presence of outliers by introducing the RGP-*t* model, which is motivated by recent advances in outlier-robust GP-based modeling. Our proposed approach incorporates the robust Student-*t* treatment of the GP-RLARX model to the multi-layer structure of recurrent GPs. A modified variational approach, REVARB-*t*, enable us to perform robust inference within such powerful modeling framework.

We comprehensively evaluated the proposed solution with computational experiments on several artificial benchmarks and datasets related to process industry systems. The flexible and resilient structure of the RGP-*t* model was responsible for its impressive performance in almost all scenarios, even with large amounts of outliers. We also analyzed how the variational approach can be applied to detect outliers in the estimation samples and avoid their spurious effect on learning and prediction.

For further work we expect to adapt the RGP-*t* model and the REVARB-*t* framework in order to perform robust identification within an online learning setup, where learning is performed "on-the-fly" as the estimation samples arrives. Another interesting application of our approach is within control techniques, such as the Model Predictive Control (MPC) methodology, where the robustness and Bayesian treatment of the RGP-*t* model could be exploited and applied to several scenarios where the assumption of Gaussian noise is not reasonable.

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References

[1] O. Nelles, Nonlinear System Identification: From Classical Approaches to Neural Networks and Fuzzy Models, Springer Science & Business Media, 2013.
[2] W. Xu, E.-W. Bai, M. Cho, System identification in the presence of outliers and random noises: a compressed sensing approach, Automatica 50 (11) (2014) 2905–2911.
[3] V. Peterka, Bayesian approach to system identification, in: Trends and Progress in System identification, vol. 1, 1981, pp. 239–304.
[4] C. Rasmussen, C. Williams, Gaussian Processes for Machine Learning, 1st edition, MIT Press, 2006.
[5] G. Matheron, The intrinsic random functions and their applications, Adv. Appl. Prob. (1973) 439–468.

[6] M.L. Stein, Interpolation of Spatial Data: Some Theory for Kriging, Springer Science & Business Media, 2012.
[7] C.K. Williams, C.E. Rasmussen, Gaussian processes for regression, in: NIPS, MIT Press, 1996, pp. 514–520.
[8] C.E. Rasmussen, Evaluation of Gaussian processes and other methods for non-linear regression (Ph.D. Thesis), University of Toronto, Toronto, Canada, 1996.
[9] R. Murray-Smith, T.A. Johansen, R. Shorten, On transient dynamics, off-equilibrium behaviour and identification in blended multiple model structures, in: Proceedings of the European Control Conference (ECC'99), Karlsruhe, Germany, 1999, pp. 3569–3574.
[10] K. Ažman, J. Kocijan, Dynamical systems identification using Gaussian process models with incorporated local models, Eng. Appl. Artif. Intell. 24 (2) (2011) 398–408.
[11] E. Solak, R. Murray-Smith, W.E. Leithead, D.J. Leith, C.E. Rasmussen, Derivative observations in Gaussian process models of dynamic systems, Adv. Neural Inform. Process. Syst. 16.
[12] J. Kocijan, A. Girard, B. Banko, R. Murray-Smith, Dynamic systems identification with Gaussian processes, Math. Comp. Model. Dyn. Syst. 11 (4) (2005) 411–424.
[13] A. Rottmann, W. Burgard, Learning non-stationary system dynamics online using Gaussian processes, in: Pattern Recognition, Vol. 6373 of Lecture Notes in Computer Science, Springer, 2010, pp. 192–201.
[14] A. Girard, C. Rasmussen, J. Quiñonero-Candela, R. Murray-Smith, Multiple-step ahead prediction for non linear dynamic systems: a Gaussian process treatment with propagation of the uncertainty, in: Advances in Neural Information Processing Systems, MIT Press, Cambridge, MA, USA, 2003, pp. 529–536.
[15] A. Damianou, N. Lawrence, Semi-described and semi-supervised learning with Gaussian processes, in: 31st Conference on Uncertainty in Artificial Intelligence (UAI'2015), 2015.
[16] T. Chen, H. Ohlsson, L. Ljung, On the estimation of transfer functions, regularizations and Gaussian processes – revisited, Automatica 48 (8) (2012) 1525–1535.
[17] R. Frigola, Y. Chen, C. Rasmussen, Variational Gaussian process state-space models, in: Advances in Neural Information Processing Systems, MIT Press, Cambridge, MA, USA, 2014, pp. 3680–3688.
[18] A. Svensson, A. Solin, S. Särkkä, T.B. Schön, Computationally efficient Bayesian learning of Gaussian process state space models, arXiv preprint arXiv:1506.02267.
[19] R.K. Pearson, Outliers in process modeling and identification, IEEE Trans. Control Syst. Technol. 10 (1) (2002) 55–63.
[20] M. Gupta, J. Gao, C.C. Aggarwal, J. Han, Outlier detection for temporal data: a survey, IEEE Trans. Knowl. Data Eng. 26 (9) (2014) 2250–2267.
[21] P.J. Rousseeuw, A.M. Leroy, Robust Regression and Outlier Detection, vol. 589, John Wiley & Sons, 2005.
[22] P.J. Huber, Robust Statistics, Springer, 2011.
[23] A. Gelman, J.B. Carlin, H.S. Stern, D.B. Rubin, Bayesian Data Analysis, vol. 2, Chapman & Hall/CRC, Boca Raton, FL, USA, 2014.
[24] R.M. Neal, Monte Carlo Implementation of Gaussian Process Models for Bayesian Regression and Classification, Tech. Rep., University of Toronto, Dept. of Statistics, 1997.
[25] P. Jylänki, J. Vanhatalo, A. Vehtari, Robust Gaussian process regression with a student-*t* likelihood, J. Mach. Learn. Res. 12 (November) (2011) 3227–3257.
[26] B. Berger, F. Rauscher, Robust Gaussian process modelling for engine calibration, in: Proceedings of the 7th Vienna International Conference on Mathematical Modelling (MATHMOD'2012), 2012, pp. 159–164.
[27] M. Kuss, Gaussian process models for robust regression, classification, and reinforcement learning (Ph.D. Thesis), TU Darmstadt, 2006.
[28] M. Kuss, T. Pfingsten, L. Csató, C.E. Rasmussen, Approximate Inference for Robust Gaussian Process Regression, Tech. Rep., Max Planck Institute for Biological Cybernetics, Tubingen, Germany, 2005.
[29] A. Naish-Guzman, S. Holden, Robust regression with twinned Gaussian processes, in: Advances in Neural Information Processing Systems, MIT Press, Vancouver, Canada, 2008, pp. 1065–1072.
[30] O. Stegle, S.V. Fallert, D.J. MacKay, S. Brage, Gaussian process robust regression for noisy heart rate data, IEEE Trans. Biomed. Eng. 55 (9) (2008) 2143–2151.
[31] N. Kantas, A. Doucet, S.S. Singh, J. Maciejowski, N. Chopin, et al., On particle methods for parameter estimation in state-space models, Stat. Sci. 30 (3) (2015) 328–351.
[32] G. Bottegal, A.Y. Aravkin, H. Hjalmarsson, G. Pillonetto, Outlier robust system identification: a Bayesian kernel-based approach, IFAC Proc. Vol. 47 (3) (2014) 1073–1078.
[33] T.B. Schön, F. Lindsten, J. Dahlin, J. Wågberg, C.A. Naesseth, A. Svensson, L. Dai, Sequential Monte Carlo methods for system identification, in: Proceedings of the 17th IFAC Symposium on System Identification (SYSID), vol. 48, Elsevier, Beijing, China, 2015, pp. 775–786.
[34] M.I. Jordan, Z. Ghahramani, T.S. Jaakkola, L.K. Saul, An introduction to variational methods for graphical models, Mach. Learn. 37 (2) (1999) 183–233.
[35] M.J. Wainwright, M.I. Jordan, et al., Graphical models, exponential families, and variational inference, Found. Trends Mach. Learn. 1 (1–2) (2008) 1–305.
[36] D.M. Blei, A. Kucukelbir, J.D. McAuliffe, Variational inference: a review for statisticians, arXiv preprint arXiv:1601.00670.
[37] T.P. Minka, Expectation propagation for approximate Bayesian inference, in: Proceedings of the 17th Conference on Uncertainty in Artificial Intelligence (UAI'01), Morgan Kaufmann, 2001, pp. 362–369.
[38] A. Gelman, A. Vehtari, P. Jylänki, C. Robert, N. Chopin, J.P. Cunningham, Expectation propagation as a way of life, arXiv preprint arXiv:1412.4869.
[39] R. Gopaluni, A particle filter approach to identification of nonlinear processes under missing observations, Can. J. Chem. Eng. 86 (6) (2008) 1081–1092.
[40] R. Frigola, F. Lindsten, T.B. Schön, C.E. Rasmussen, Identification of Gaussian process state-space models with particle stochastic approximation EM, arXiv preprint arXiv:1312.4852.
[41] P. Green, Bayesian system identification of a nonlinear dynamical system using a novel variant of simulated annealing, Mech. Syst. Signal Process. 52 (2015) 133–146.
[42] D. Barber, Bayesian Reasoning and Machine Learning, Cambridge University Press, Cambridge, UK, 2012.
[43] K.P. Murphy, Machine Learning: A Probabilistic Perspective, MIT Press, Cambridge, MA, USA, 2012.
[44] M.K. Titsias, Variational learning of inducing variables in sparse Gaussian processes, in: International Conference on Artificial Intelligence and Statistics (AISTATS'2009), 2009, pp. 567–574.
[45] M.K. Titsias, N.D. Lawrence, Bayesian Gaussian process latent variable model, in: International Conference on Artificial Intelligence and Statistics, 2010, pp. 844–851.
[46] A. Damianou, Deep Gaussian processes and variational propagation of uncertainty (Ph.D. Thesis), University of Sheffield, UK, 2015.
[47] C.L.C. Mattos, J.D.A. Santos, G.A. Barreto, An empirical evaluation of robust Gaussian process models for system identification, in: Intelligent Data Engineering and Automated Learning-IDEAL 2015, Springer, 2015, pp. 172–180.
[48] C.L.C. Mattos, A. Damianou, G.A. Barreto, N.D. Lawrence, Latent autoregressive Gaussian process models for robust system identification, in: Proceedings of the 11th IFAC DYCOPS-CAB, Trondheim, Norway, 2016, pp. 1121–1126.
[49] R. Ranjan, B. Huang, A. Fatehi, Robust Gaussian process modeling using EM algorithm, J. Process Control 42 (2016) 125–136.
[50] R. Pascanu, C. Gulcehre, K. Cho, Y. Bengio, How to construct deep recurrent neural networks, in: Proceedings of the Second International Conference on Learning Representations (ICLR'2014), 2014.
[51] A. Damianou, N. Lawrence, Deep Gaussian processes, in: Proceedings of the 16th International Conference on Artificial Intelligence and Statistics (AISTATS'2013), 2013, pp. 207–215.
[52] C.L.C. Mattos, Z. Dai, A. Damianou, J. Forth, G.A. Barreto, N.D. Lawrence, Recurrent Gaussian processes, in: Proceedings of the 4th ICLR, San Juan, Puerto Rico, 2016.
[53] S.A. Billings, Nonlinear System Identification: NARMAX Methods in the Time, Frequency, and Spatio-temporal Domains, John Wiley & Sons, 2013.
[54] M.E. Tipping, N.D. Lawrence, Variational inference for student-*t* models: robust Bayesian interpolation and generalised component analysis, Neurocomputing 69 (1) (2005) 123–141.
[55] C.M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006.
[56] R Core Team, R: A Language and Environment for Statistical Computing, R Foundation for Statistical Computing, Vienna, Austria, 2016, https://www.R-project.org/.
[57] Theano Development Team, Theano: a Python framework for fast computation of mathematical expressions, arXiv e-prints abs/1605.02688. URL http://arxiv.org/abs/1605.02688.
[58] M. Abadi, P. Barham, J. Chen, Z. Chen, A. Davis, J. Dean, M. Devin, S. Ghemawat, G. Irving, M. Isard, et al., Tensorflow: a system for large-scale machine learning, in: Proceedings of the 12th USENIX Symposium on Operating Systems Design and Implementation (OSDI), Savannah, Georgia, USA, 2016.
[59] D.M.B. ed., DaISy: Database for the identification of systems, http://homes.esat.kuleuven.be/ smc/daisy/, accessed July 2016.
[60] K.S. Narendra, K. Parthasarathy, Identification and control of dynamical systems using neural networks, IEEE Trans. Neural Netw. 1 (1) (1990) 4–27.
[61] J. Campos, F.L. Lewis, R. Selmic, Backlash compensation with filtered prediction in discrete time nonlinear systems by dynamic inversion using neural networks, in: Proceedings of the 39th IEEE Conference on Decision and Control, vol. 4, 2000, pp. 3534–3540.
[62] L. Kaufman, P.J. Rousseeuw, Partitioning around medoids (program PAM), in: Finding Groups in Data: An Introduction to Cluster Analysis, 1990, pp. 68–125.
[63] M. Maechler, P. Rousseeuw, A. Struyf, M. Hubert, K. Hornik, cluster: Cluster Analysis Basics and Extensions, R package version 2.0.4 (2016).